



Review article

Energy efficient sustainable Wireless Body Area Network design using network optimization with Smart Grid and Renewable Energy Systems

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ABSTRACT

Technological advancements have made intelligent monitoring devices such as integrated circuits and wireless communications smaller, lighter, and more energy-efficient. When combined with WBAN (Wireless Body Area Network), these devices can be connected seamlessly with Smart Grid and Renewable Energy Systems. Devices of this type are compact and include a power supply, resulting in limited power systems. The body's sensors are small, and the battery cannot hold for a long time; hence, better power management strategies are required. In this paper, Energy Efficient Sustainable Network Using Network Optimization Technique (EES-NOT) is proposed to maximize the energy efficiency of the WBAN. The adaptive scheduling protocol for energy efficiency (ASPEE) is integrated with EES-NOT to enhance the network's reliability and Quality of Service (QoS). Furthermore, a secured routing protocol for energy efficiency (SRPEE) reduces network congestion and provides optimal route selection. The simulation results show that the suggested EES-NOT model enhances the throughput, network lifetime, transmission overhead, latency, and transmission rate with an achievement of 96.3% compared to other existing approaches.

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1. Overview of the Energy Efficient Sustainable Wireless Body Area Network Design

People today lead unhealthy lifestyles due to poor choices, exercise habits, and insufficient release from extreme stress (Preeth et al., 2018), contributing to chronic disease's beginning and development (Gunasekaran and Narayanasamy, 2018). As a result, Medical practices and preventative medicine are critical nowadays (Gao et al., 2020b). Non-communicable diseases are the leading cause of mortality globally because patients do not recognize or experience symptoms early in the disease (Do et al., 2020). Healthcare systems must be widely available, reasonably priced, and capable of diagnosing illnesses early on (Saravanan et al., 2011).

With the advancements in technology, the miniaturization of sensor devices has made Wireless Body Area Network (WBAN) an attractive option because of its low cost (Karthikeyan et al., 2020). Sensors in WBAN can communicate through a wireless medium because they are small, low-powered, and lightweight with Smart Grid and Renewable Energy Systems (Mohanty et al., 2020). The heart rate, respiratory rate, and other human physiological factors can be quantified with the help of WBANs (Li et al., 2020). Patients no longer have to visit the hospital frequently for check-ups because of continuous and remote health monitoring (Amudha and Narayanasamy, 2018). Thus, WBAN allows its customers to choose their preferred location while saving on healthcare expenses (Gao et al., 2020a).

Due to the limited sensors' recharging batteries, changeable network topology; strict QoS requirements (Nguyen et al., 2016); and fluctuating file generation and storage rates, WBAN technology faces numerous challenges (Shakeel et al., 2020). The durability of WBANs is dependent on energy consumption, which makes them one of the most common problems in the network (Lin et al., 2021). With this ability, addressing these issues is essential if the WBAN is adequately implemented (Islam et al., 2019).

Many different approaches to optimizing a network's performance across other network domains (Chen et al., 2021). Network administrators can use network optimization techniques to monitor, manage, and improve key performance indicators to benefit all network users (Wang et al., 2021). By making optimal use of network system resources (Alzahrani and Almotairi, 2019). Network Optimization keeps track of key performance indicators (Khattak et al., 2019), and it reports them in real-time to give network managers a head start on network management (Al Aghbari et al., 2019). By utilizing optimization, network managers can foresee the effects of architecture changes on the network before implementation (Kumar et al., 2020). Ultimately, these advantages lead to improved network performance (Cicioğlu and Çalhan, 2019).

The information contained in a person's vital signs is strictly confidential. As a result, mechanisms for ensuring security and privacy are critical in WBAN (Mu et al., 2019). Data packets' confidentiality and authenticity depend on the security and privacy mechanisms integrated with the routing protocols (Javed et al., 2019). Therefore, the need for a more energy-efficient, secure, reliable, and throughput-maximizing approach to route selection is critical.

In contrast, the fading becomes the primary source of unreliability with a Time Division Multiple Access (TDMA) strategy because the nodes can function according to a predetermined schedule following contention-free channel access (Awan et al., 2019). When using TDMA-based network connectivity, the nodes'

scheduled duty cycles help forecast channel state even while conserving energy. Wi-Fi body area networks (WBANs) are becoming increasingly important in the design of algorithms for pervasive computing because of their advanced development and benefits for healthcare applications, particularly for early-stage disease monitoring and diagnosis and analysis on patients with the most critical and various disease stages, including the presence or absence of tumors, metabolic syndrome, chronic obstructive pulmonary disease, and cardiovascular disease.

The major contributions of this research include,

1. Designing an Energy Efficient Sustainable Network using Network Optimization Technique (EES-NOT) to enhance the energy efficiency and performance of the body area network.
2. Network performance evaluation based on the EES-NOT technique with Smart Grid and Renewable Energy Systems.
3. Analysis of the outcomes on the proposed technique with the existing methods.

The remainder of the study is structured as follows: Section 1 has given a brief introduction to the WBAN. Section 2 discusses some of the current approaches regarding the WBAN. Detailed explanations of the proposed protocols are provided in Section 3. Section 4 discusses the numerical simulations and results of the tests conducted to determine how well the strategies proposed work. Finally, Section 5 wraps up the paper's contribution and discusses its future directions.

2. Background study

This section discusses the several research works carried out on providing energy efficiency cellular networks in the human body and the drawbacks of the research. Further, it identifies the challenges to be addressed.

Sun et al. (2020) proposed an intelligent quality of service-based energy efficiency (IQoS-EE) method to improve network performance, features such as on-time and on-budget delivery, and efficiency. The sampling frequency, exposure time, and high bandwidth are all considered when determining the sampled timeslot. The results quantitatively show that the data frame size impacts the overall probability of delivery, taking the node's channel condition. Scheme enhanced network performance as well as delivery probability and time.

Ahmed et al. (2020) focused on developing an efficiently optimal congestion control. The temperature-aware routing algorithm data frame size impacts the overall probability of delivery (EOCC-TARA) has congestion control built-in for maximum efficiency. Priority was given to forwarding nodes low in energy and low in temperature for route formation in the proposed EOCC-TARA routing algorithm. For EMSMO-based networks, the best routing paths were determined using a cost model that included residual energy with connection dependability, path loss, and queue length. According to research, evidence supports the suggested EOCC-TARA routing algorithm outperforms more traditional models for SDN-based WBAN in simulations. Wang et al. (2020) suggested resource allocation schemes for solving the energy congestion method based on time handling (ECM-TH) problems. In the absence of a power source, a novel GD/BS hybrid method was proposed that achieves the performance is identical to that achieved using the inner point technique while requiring significantly less computational effort. Non-convex ECM-TH problems are first converted to convexes, and then the battery Lagrange dual subgradient approach was used to address the problem. It is less likely that a battery-assisted system will go down than a system that does not use batteries.

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Amjad et al. (2019) formulated an optimal solution for EE maximizing suboptimal solution network (EEMSSN); maximizing the efficient usage of energy-critical nodes is made possible by EE. A suboptimal solution has been proposed that comes close to the optimal solution while requiring fewer issues. The modeling results show that the proposed solutions work and that the network's energy efficiency is superior.

Bhatia and Sood (2020) depicted an IoT-based Quantum Computing-inspired Optimization (IoT-QCiO) mechanism to diminish the amount of available space immediately. The authors have specified two approaches: based on the parameters of Sensor in Vicinity (SIV) and Optimal Sensor Space (OSS). Under the current scenario, the proposed model proved extremely successful and efficient in optimizing the IoT sensor space for precise data collection in a time-sensitive way.

Hasan et al. (2019) presented a detailed survey on WBAN that reviewed the current research and challenges in WBANs. Various WBAN technologies were studied, and their characteristics have been highlighted. The survey results showed that incorporating new technologies such as SDN, EH, and blockchain WBAN can transform healthcare.

Navya and Deepalakshmi (2021) framed energy-efficient computing technique (EECT) where Biosensors and ultra-advanced nodes are used to transmit emergency data in high-stress situations. Hence to reduce energy consumption, threshold values are applied to the sensed data, and the most efficient path is found using the cost value metric to find the most efficient route. Transceiver characteristics are selected based on energy constraints and to maintain a level of balance throughout the network's lifetime.

The survey implies that the establishment of WBAN faces many challenges like reliability, energy efficiency, implementation cost, and more. This research provides an energy-efficient sustainable Wireless Body Area Network that overcomes the optimized network's power management issues. The following section discusses the proposed model briefly.

3. Energy Efficient Sustainable Network using Network Optimization Technique

The proposed systems aim to provide a highly secured, energy-efficient, sustainable wireless body area network. The objective is achieved by implementing two protocols on the data and scheduling using Smart Grid and Renewable Energy Systems.

Fig. 1 shows the general architecture of a Wireless Body Area Network. The sensors are connected to the patients, and each can communicate with the body coordinator to ensure intra WBAN communication using Smart Grid and Renewable Energy Systems. The coordinator communicates with the Personal Device Assistant, which transfers the data collected to the medical server for recording purposes. The patient's physician receives a copy of the information. The physician can retrieve previous records from the medical server and take decisions accordingly. The physician can communicate to the patient through the PDA for changing the prescriptions or providing any remote assistance. In emergency time, the alert message will be sent to the physician to commute the patient with the help of an ambulance.

Using WPAN wireless technologies as gateways, a WBAN system can extend its range. It is possible to connect wearable devices to the internet via gateway devices. Patients' locations are irrelevant because medical professionals can access patient data via the internet.

3.1. Secured Routing Protocol for Energy Efficiency (SRPEE)

A Secured Routing Protocol for Energy Efficiency is being developed to increase network energy efficiency while improving network performance and security. The Figure reveals the flow diagram of the proposed SRPEE. The protocol initializes by sending hello packets to the sensor nodes. Once the receiver node receives the hello packet, the connection is established between the nodes and ready for data transfer. BC compresses the data using the Huff Mann Coding compression technique and data encryption using the RSA encryption method using Smart Grid and Renewable Energy Systems. The PSO routing algorithm makes the optimal node selection. The proposed method is implemented as a function of the coordinator.

Hello, packets are sent to the sensor nodes to get the protocol up and run. Hello, packets are exchanged between two nodes when the receiver node receives the message. BC uses Huff Mann Coding compression and RSA encryption to compress and secure data. The PSO routing algorithm selects the optimal node. The coordinator is responsible for implementing the method proposed here.

3.1.1. Initialization phase

At regular intervals, BC sends hello packets, and the other sensors (N_{node}) reply with a hello packet after receiving hello packets from BC if it lies between the Source (S_{node}) and Destination (D_{node}). The S_{node} adds N_{node} Information on the routing table. Fig. 2 depicts the flow diagram of the initialization phase and the minimal cost calculation for the next hop.

BC sends hello packets at regular intervals, and other sensors (N_{node}) respond with hello packets if they are between the source (S_{node}) and destination (N_{node}) (D_{node}). The routing table gains N_{node} Information as a result of the S_{node} . A minimal cost calculation is performed during the initialization phase to determine where to begin and where to end the hop.

3.1.2. Data compression

The sensor continuously monitors the patient, requiring additional storage, resulting in a storage capacity issue for each sensor node. The problems mentioned above can be resolved by data compression. Compression using Huffman is lossless, yet the non-uniform method of compressing large amounts of data efficiently is given in Fig. 3. It improves network security by compressing and encrypting data using Smart Grid and Renewable Energy Systems. It transmits data with a smaller packet size since it uses fewer bits. More bits mean less energy usage and less storage space in sensor nodes; reducing the number of bits is good. With this, data packet transfer speeds will be increased, redundant sensing data will be eradicated, space and time wasted will not be destroyed, and storage hardware expenses will be reduced while increasing network capacity. The Efficiency of the compression is calculated as Eq. (1).

$$\eta = \frac{H(z)}{L(z)} \times 100 \quad 0 \leq \eta \leq 100 \quad (1)$$

where $H(z)$ is Entropy (S) calculated by Eq. (2)

$$H(z) = - \sum_{i=1}^n P(m_i) \log P(m_i) \quad (2)$$

$$L(z) = \sum_{i=1}^n P(m_i) L(m_i) \quad (3)$$

$P(m_i)$ is the probability of sensor data and $L(m_i)$ is the length of the data is obtained using Eq. (3).

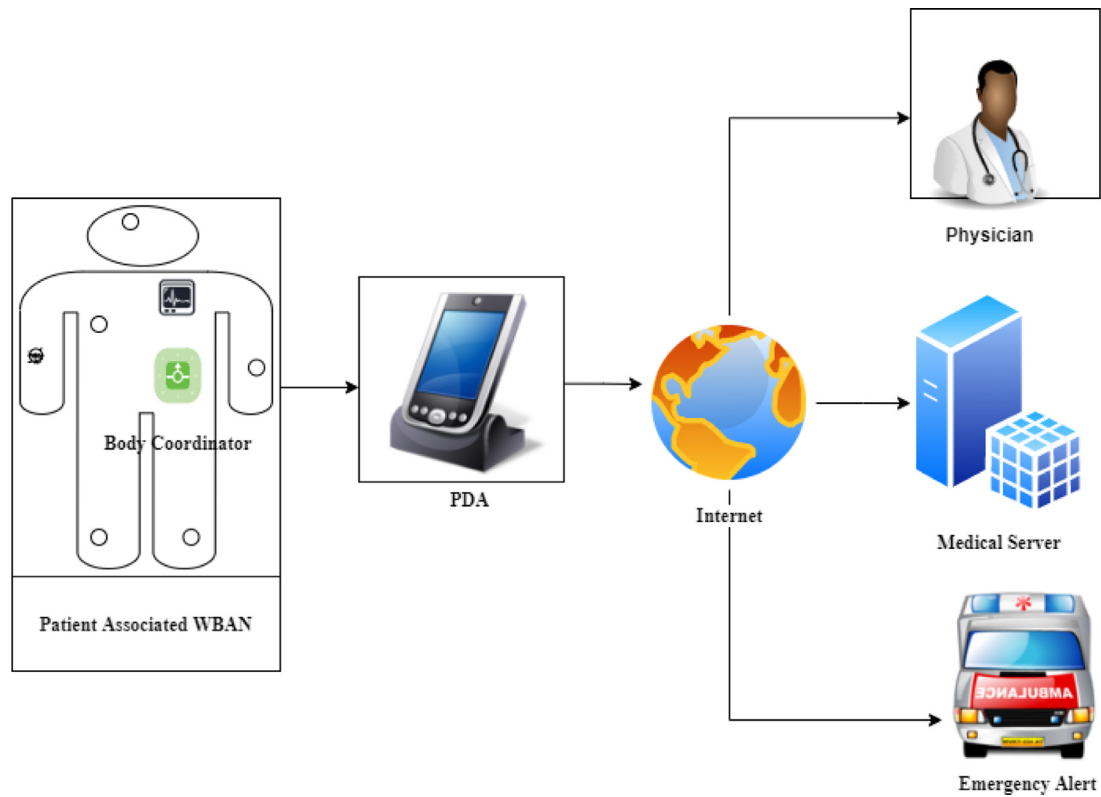


Fig. 1. WBAN architecture.



Fig. 2. Flow diagram of SRPEE.

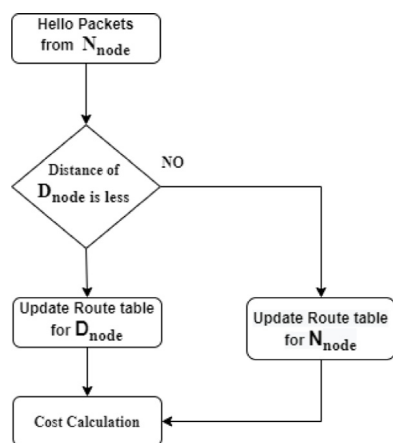


Fig. 3. Flowchart for minimal cost calculation.

3.1.3. Data encryption

The proposed method uses RSA, an asymmetric cryptographic technique, for enhancing data security and privacy. The encrypted

data is given by Eq. (4).

$$E_{data} = P^e \bmod m \quad (4)$$

The decrypted data is given by Eq. (5)

$$P_{data} = E^d \bmod m \quad (5)$$

E_{data} = Encrypted data to be transmitted, P_{data} = Patient data collected from the sensor, (m, e) = RSA Public key, d = Private key illustrated in Eq. (6).

$$ed = 1 \bmod (p-1)(q-1), \text{ where } p * q = m \quad (6)$$

Transforming plaintext (data that has not been encrypted) into ciphertext is the goal of data encryption (encrypted). Users can access encrypted and decrypted data, respectively, using an encryption key and a decryption key.

Restoring encrypted data to its original form is known as decryption. While being shared or transmitted, information is encrypted from its original format (known as plaintext) into an unreadable format (known as ciphertext).

3.1.4. Route selection

Efforts are being made to determine the most energy-efficient way to send data packets while maintaining high throughput. The minimal Cost (MC_h) for next-hop is given by Eq. (7), and

the Distance from source to destination ($D_{S,D}$) is calculated as per Eq. (8).

$$MC_h = w_1 \left| \frac{D_{S,N}}{D_{S,D}} \right| + w_2 \left| \frac{L_{pdata} - L_{edata}}{L_{pdata}} \right| + w_3 \left| \frac{E_0 - RE_N}{E_0} \right| \quad (7)$$

where $\frac{D_{S,N}}{D_{S,D}}$ is the distance from source to destination, L_{pdata} is the length of packet data, L_{edata} is the length of encrypted data. The router compares routing table entries to determine a given packet's next hop (by adding it with the netmask and comparing it to the network address). The packet will be forwarded to the first next-hop that matches the IP address.

$$D_{S,D} = \sqrt{(x_D - x_S)^2 + (y_D - y_S)^2} \quad (8)$$

where x_D, x_{S_i} x coordinates of source and destination y_D, y_S are y coordinates of origin and destination

This equation uses the distance formula to calculate the distance taken between 2 hops.

The transmitted energy and received energy are calculated using Eqs. (9) and (10),

$$TE = kE_{Tx} + kE_{amp}D_{i,j} \quad (9)$$

$$RE = kE_{Rxb} \quad (10)$$

The increased energy efficiency of the proposed model is better than the existing models.

3.2. Adaptive Scheduling Protocol for Energy Efficiency

As traffic loads change, the proposed ASPEE adapts the sleep duration of nodes to minimize energy consumption using Smart Grid and Renewable Energy Systems. A threshold is generated randomly to prevent repeated switching between active and sleep states to save energy. Node sleep duration is shortened to reduce switching energy waste when the network's traffic volume increases. The suggested approach reduces the likelihood of packet collisions while reducing the time needed to monitor and control network activity. With the advanced technique, difficulties caused by active and inactive nodes can be reduced to a minimum.

The proposed protocol works based on request and grant policy using Smart Grid and Renewable Energy Systems. Keeping all the nodes active simultaneously may consume more power, and keeping the idle node in sleep mode will cause deep fading. Therefore, the Body coordinator acts as the central node that handles the requests from the other nodes.

The node ready with the data first sends the request the data. Depending on the length of the sensor data, the BC allocated the time slot. Sensors with short length data that will be granted access first are framed in Fig. 4. The sensors with critical data like Electro Cardio Gram (ECG) will be given access immediately by releasing the current key using Smart Grid and Renewable Energy Systems

The network's nodes are distributed randomly amongst themselves in the total number N . Nodes could identify other nodes in their communication range thanks to the uniform transmission range (Tr).

$G = (V, E)$, where E denotes Distance (edges) and V represents the collection of nodes (vertices). As long as the distance between two nodes is less than the Euclidean distance, they are considered nearby. The Distance between two nodes is determined by their proximity to one other and Tr .

As traffic loads change, the proposed ASPEE adapts the sleep duration of nodes to minimize energy consumption. A cutoff point based on various factors changed to reduce energy consumption to avoid frequent transitions in Fig. 5. Node sleep duration is shortened to reduce switching energy waste when traffic

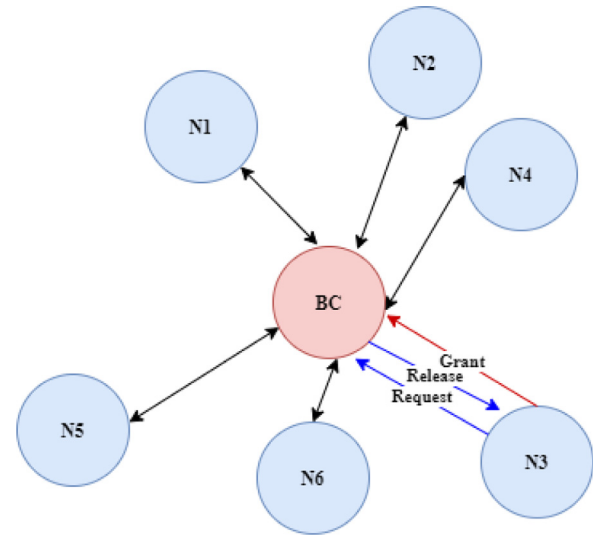


Fig. 4. Adaptive scheduling method.

load grows in the network using Smart Grid and Renewable Energy Systems. The suggested approach reduces the likelihood of packet collisions while reducing the time needed to monitor and control network activity. With the suggested technique, difficulties caused by active and inactive nodes can be reduced to a minimum.

The network's nodes are distributed randomly amongst themselves in the total number N . All nodes had the same transmission range (Tr), making it easy to identify the other nodes in their communication range.

Symbol

$E = \{d1, d2, d3, d4, \dots, dn\}$

$V = \{v1, v2, v3, v4, \dots, vn\}$

$N1$ and $N2$

$H1$

$H2$

τ_t^i

τ_{ack}^i

$Hopdelay_{onehop}$

τ_s

$P(m_i)$

$L(m_i)$

E_{data}

Description

Edges

The network of connections

Nodes

Single Hop

two-hop

Sending time

Acknowledge the time of i th packet.

Hop time

Samples

Probability of sensor data

Length of the data

Encrypted data

$G = (V, E)$; where, $E = \{d1, d2, d3, d4, \dots, dn\}$ represents the distance (edges) and $V = \{v1, v2, v3, v4, \dots, vn\}$ illustrates the network of connections. As long as the distance between nodes u and v is smaller than the Euclidean distance, they are considered nearby nodes in Eq. (11). Where the edge $e = (N1, N2) \in E$ and $v = (N1, N2) \in V$.

$$E = \{(N1, N2) \in V.V [D_{EU}(N1, N2) \leq Tr]\} \quad (11)$$

If both $N1$ and $N2$ have a direct link, then nodes, $N1$ and $N2$, a single hop ($H1$) each other. The relationship between $(N1, N2)$ and E can be seen in equation ($E H1$) (12).

$N1$ and $N2$ are considered to be in a two-hop neighborhood ($H2$) when they do not have a direct link $(N1, N2) \in E$ between them, instead have a link $(N1, N3), (N3, N2) \in E$. $N1$ and $N2$ are equal to $E H2$. there is a connection If n is more than or equal to three, the answer is (3)

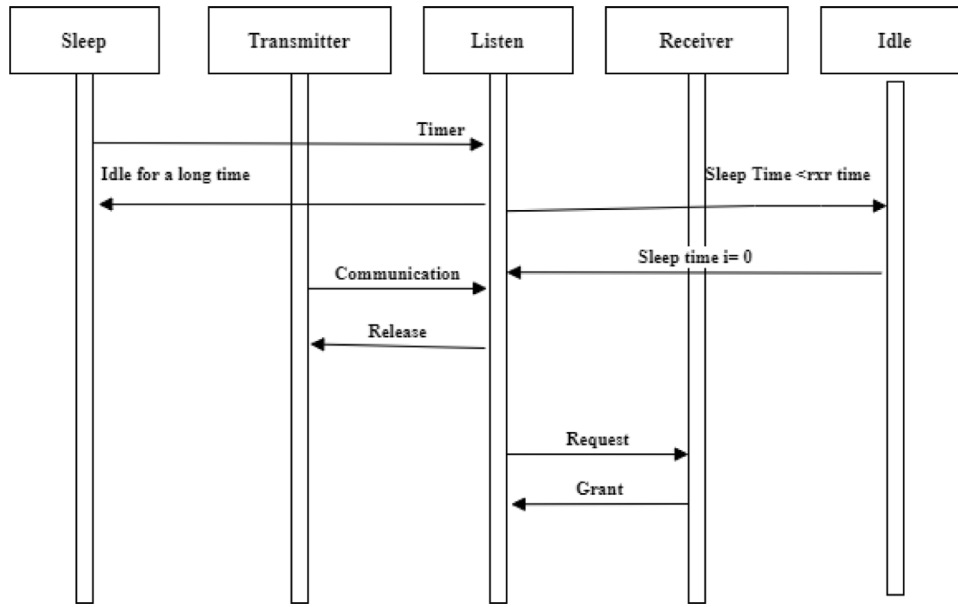


Fig. 5. UML diagram of adaptive scheduling protocol.

Eq. (12) can be used to determine the waiting time at node N1.

$$\tau_{u,v}^w \leq \sum_{i=0}^n (\tau_t^i + \tau_{ack}^i) \quad (12)$$

where τ_t^i is the sending time and τ_{ack}^i is the acknowledge time of i th packet.

The hop time is calculated with Eq. (13)

$$\text{Hop delay}_{\text{one hop}} = (\tau_{N1,N2}^{N3} + \tau_{data}) \quad (13)$$

τ_s Samples each channel after the sleep period has ended. To prevent this, the node will enter the sleep state once more after receiving an unintended packet for a while, τ_a

The Adjusting sleep time is calculated by Eq. (14).

$$\tau_t = \tau_1 * n \quad (14)$$

If n indicates the total number of received packets, and τ_1 represents the receiving time for a single data packet, the τ_t indicates how much time has passed since the beginning of the program.

$$\text{The maximum Sleep time } \tau_s = \tau_{sleep}^M \quad (15)$$

Utilizing system resources efficiently and sharing scheduling information allows the ASPEE to cut control costs obtained using Eq. (15). Nodes can concurrently reserve a slot with the suggested approach without adding control overhead in a dynamic context. The ASPEE automatically modifies the sleep period of nodes based on the network traffic load.

Health care facilities use a private storage facility that includes indexing and retrieval functions. Health care centers and end-users here the human body rely on a processing server to handle simultaneous tasks. Fig. 6 depicts how the operations of a health care center and its end-users differ. It is the responsibility of the end-users to request and retrieve information from the health care center. Indexing and retrieval are the framework's core functions, and they work in concert with concurrency to improve communication between healthcare facilities and their customers in optimization.

$$\text{Hop delay}_{\text{one hop}} = (\tau_{N1,N2}^{N3} + \tau_1 * n) \quad (16)$$

$$\rho(H) = \frac{1}{1 - \sum_{i=1}^R \text{Hop delay}} \quad (17)$$

Table 1

Combining nodes with other methods.

Number of nodes	IQoS-EE	EOCC	TARA	ECM-TH	EEMSSN	EES-NOT
5	19.78	21.21	24.54	18.76	28.76	24.1
10	21.78	12.65	45.36	73.56	32.45	35.23
15	18.65	26.33	53.66	48.23	46.76	55.13
20	32.15	18.98	35.97	53.26	30.12	78.43
25	28.56	36.78	61.78	31.03	51.34	77.11
30	45.89	31.87	49.74	45.69	67.87	74.2
35	38.27	36.21	51.23	50.89	50.65	87.15
40	48	46.3	61.23	58.89	60.8	88.14
45	58	54.3	71.5	63.89	70.65	86.15
50	68.26	79.21	81.46	69.31	80.12	96.3

where Eqs. (16) and (17) represents the Hop delay and efficiency denoted by the parameter $\rho(H)$. Hence the efficiency increases if hop delay decreases.

The proposed EES-NOT technique is efficiently designed with SPREE and ASPEE. The efficiently reduces the energy consumption with low transmission power and provides High security and accuracy with low latency and high packet delivery ratio using Smart Grid and Renewable Energy Systems.

4. Results and analysis

The proposed EES-NOT for Wireless Body Area Network is modeled and assessed using the recommended experimental setup. The performance of the suggested EES-NOT is compared with the existing methodologies like IQoS-EE, EOCC TARA, ECM-TH, and EEMSSN using Smart Grid and Renewable Energy Systems. It is determined how well the network performs based on the parametric such as Energy Consumption, Hop Count, Packet Delay Ratio, Network Lifetime Ratio, Transmission rate, and throughput.

Table 1 indicates the technique for evaluating the effectiveness of the current models such as IQoS-EE, EOCC TARA, ECM-TH, and EEMSSN, and the proposed plan of EES-NOT. The transmission efficiency and their related approaches in EES-NOT are described.

4.1. Energy consumption analysis

Fig. 7 shows the simulation results for variation in energy consumption ratio concerning the constant increase in nodes. The

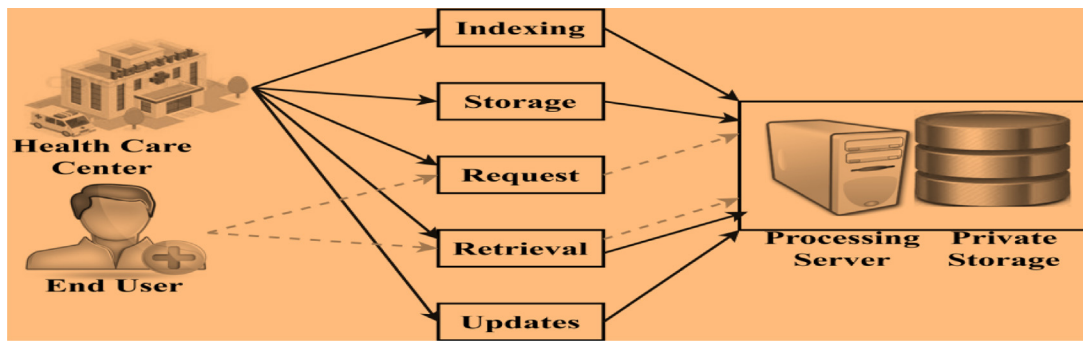


Fig. 6. Functions of Wban.

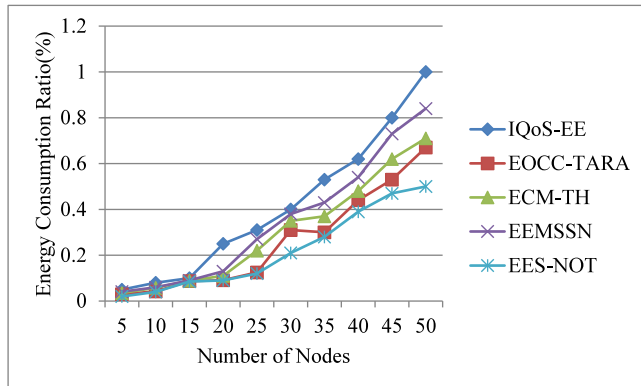


Fig. 7. Energy consumed ratio analysis.

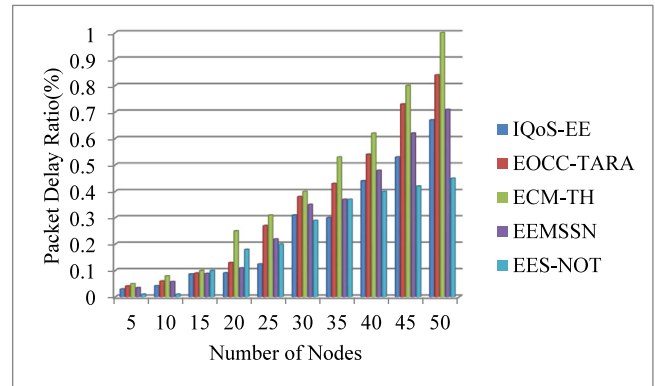


Fig. 9. Packet delay ratio analysis.

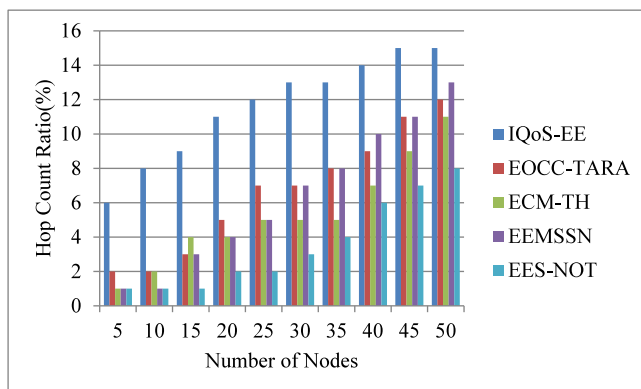


Fig. 8. Hop count ratio analysis.

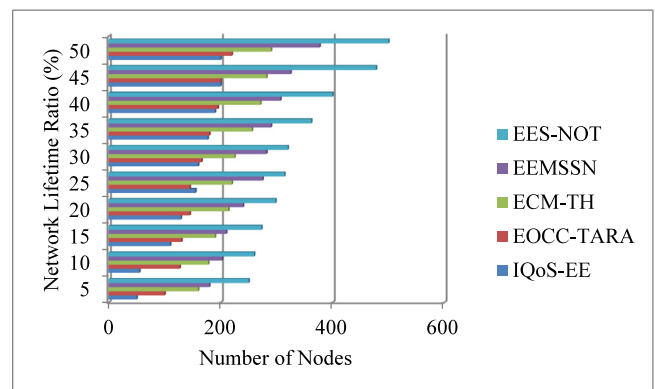


Fig. 10. Network reliability to lifetime analysis.

proposed technique uses an optimization method to select the next-hop node using Smart Grid and Renewable Energy Systems. The routing consumes energy, and hence the minimal cost in Eq. (7) is considered to be calculated as the energy consumption ratio. The graph presented displays that as the number of sensor nodes grows, the new technique EES-NOT shows a considerable decrease in energy consumption.

4.2. Hop count ratio analysis

Counting the network devices that data travels through to its final destination is the hop count. The main aim of the SRPEE protocol is to find the minimal cost distance from the source to the destination, which can reduce the consumption of energy wasted in reaching the destination with the available path using Smart Grid and Renewable Energy Systems. Fig. 8 illustrates the

variations of hop count ratio with the change in the number of nodes. From the graph, it is inferred that integrating the newly proposed protocol reduces the hop count with an efficient route selection procedure compared to the existing techniques.

4.3. Packet delay ratio analysis

Network latency refers to any delay in data communication across a network, known as packet delay. Fig. 9 demonstrates the performance of the proposed EES-NOT in terms of Packet Delay Ratio with an increasing number of nodes. The variation in the graph depicts that the newly designed technique has reduced delay in the packet delivery from the source to destination as it takes the shortest path with the ASPEE.

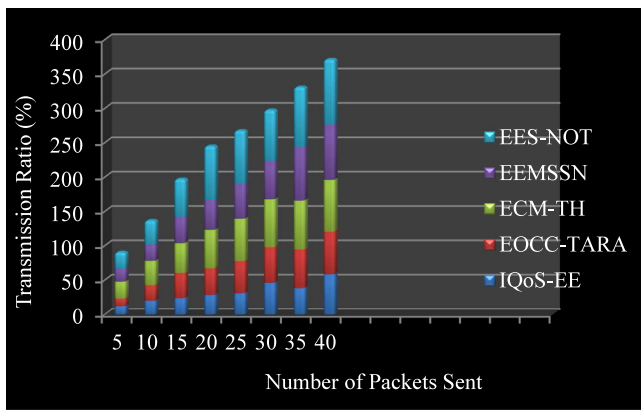


Fig. 11. Transmission ratio analysis.

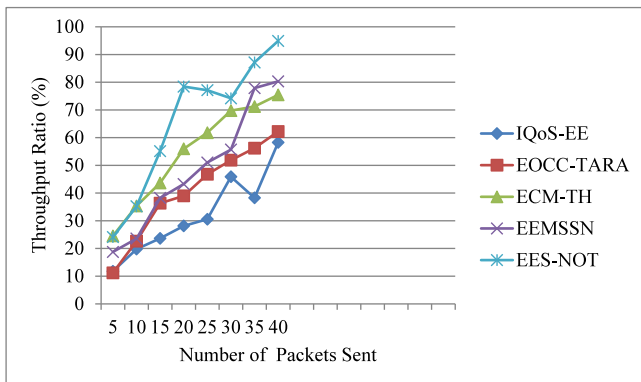


Fig. 12. Throughput analysis.

4.4. Network lifetime ratio analysis

Network Lifetime refers to how long sensors put in a network may continue to observe the phenomena of interest. Fig. 10 compares the network lifetime achieved by various conventional methods existing with the performance of the proposed technique EES-NOT. The results instantiate that the ASPEE adopted in the research has enhanced the network lifetime with nodes.

4.5. Transmission ratio analysis

The new protocols incorporated in the research have increased the packets' transmission rate by increasing the number of nodes given in Fig. 11. The transmission rate is determined as the measured number of data units delivered at the measurement interval for each measurement period. The graph presented below exhibits the transmission rate given by the surveyed techniques and the proposed method.

The number of nodes shown in Fig. 10 has increased due to the new protocols incorporated into the research. The transmission rate is calculated by dividing the number of data units delivered by measurement periods.

4.6. Throughput analysis

The number of packets successfully received at the destination node per unit of time is referred to as the throughput. The patient data are to be analyzed with the utmost care; they may suggest emergency alert situations, as illustrated in Fig. 12. The proposed approach embellishes a high performance compared to the other protocols with the high throughput achieved by the compression

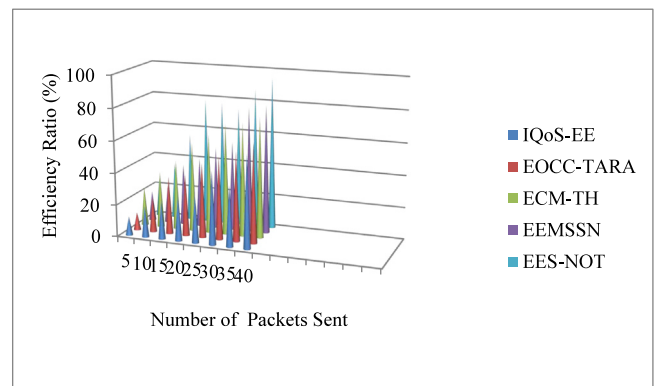


Fig. 13. Efficiency analysis.

technique and the optimal cost calculation using Smart Grid and Renewable Energy Systems. The Figure shows the throughput variations with an increase in the number of packets.

4.7. Efficiency analysis

Efficiency refers to the number of packets successfully received at the destination node per unit of time. The patient data must be analyzed with extreme caution, as shown in Fig. 13, because they may indicate situations requiring an emergency alert. Using Smart Grid and Renewable Energy Systems, the proposed approach enhances the high performance compared to other protocols by increasing efficiency through compression. As the number of packets increases, you can see in the graph how efficiency changes.

The result analysis depicts that the proposed EES-NOT design for WBAN Smart Grid and Renewable Energy Systems improves overall network performance by providing improved throughput, packet delivery rate, network lifetime, and reduced energy consumption rate, latency, and Hop Count with an achievement of 96.3% compared to the existing IQoS-EE, EOCC-TARA, ECM-TH, EEMSSN techniques.

5. End notes

The purpose of this paper is to present an Energy Efficient Sustainable Network Using Network Optimization Technique (EES-NOT) based WBAN that enhances the performance and energy efficiency of the network using Smart Grid and Renewable Energy Systems. Two new protocols, SRPEE and ASPEE, are introduced to achieve this objective. Secured Routing Protocol for Energy Efficiency (SRPEE) compresses, encrypts the data collected, and selects the optimal next hop node providing high data security and less network congestion. Adaptive Scheduling Protocol for Energy Efficiency (ASPEE) dynamically allocates slots to nodes, avoiding the fading of a node from the network. The effectiveness of the new procedures has been assessed and compared to that of the current ones. The results show that the new system has high network performance, increased energy efficiency, throughput, transmission rate, and reduced packet dropping rate and latency with an achievement. Future work will expand the network performance by incorporating new technologies like big data analytics, machine learning, deep learning, and blockchain technology to make the system interactive.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

The data supporting the findings of this work are available within the article.

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